

SAP LABS INDIA– INTERVIEW EXPERIENCE

About the Company:

SAP (short for Systems, Applications, and Products in Data Processing) is one of the biggest names in enterprise software, helping businesses around the world run smarter, whether it's managing their supply chains, handling finances, or making sense of their data.

SAP Labs India is a key innovation and R&D hub for SAP, with a global presence in five cities: Bengaluru, Gurgaon, Pune, Mumbai, and Hyderabad. Bengaluru hosts the largest campus, serving as a crucial center for driving AI solutions and enterprise technology development for global customers. The labs focus on a customer-centric and agile approach to innovation, covering user experience, business models, and cutting-edge technology.

Role Offered: Developer Associate

Selection Process:

1. Online Test
2. Technical Round 1
3. Technical Round 2
4. Techno-managerial Round
5. HR Round

For me, only four rounds were held including OA. I think they might have combined Tech 2 and Techno-Managerial. I spent almost the whole day waiting for round one, and once it started, the rounds blasted off like a rocket.

Online Assessment(@CUIC):

About 280 students made it to the online assessment round. The OA happened in the IT lab and ran for a solid **100 minutes** (yep, 1 hour 40 mins — long enough to question your life choices). The test was on Codility, with both video *and* audio proctoring.

We got **2** DSA questions each, and everyone had their own mystery set with different kinds of questions. One was supposed to be “easy”, and the other might fit in the “medium” zone. But I'd say, I got a 1+1 offer!! both turned out to be mediums.

Questions asked for me:

1) **Problem:** Cover the Holes with Wood

You are given a table with holes at certain positions. Your task is to cover all the holes using **exactly two wooden planks** of the **same length**. You have woods of infinite size and also the table has infinite length, but you want to minimize the wooden plank size.

- An integer array **holes[]** representing the positions of holes on the table.
- $1 \leq \text{holes.length} \leq 10^5$, $0 \leq \text{holes}[i] \leq 10^9$ (unique)

Eg: holes = [1, 3, 4, 7, 9], **Output:** 5

Explanation: One plank of length 5 can cover holes at positions [1,3,4], and other plank for the remaining holes.

My thoughts: It is basically a medium-easy qns which can be solved using binary search.

2) **Problem:** Graph Disappearance Time

You are given a graph which may or may not be connected. Each second, any node with **at most one edge** disappears, and all edges connected to it also disappear. Your task is to determine the **minimum time** required for the entire graph to disappear.

Eg:

Input: n = 5, edges = [[1,2],[2,3],[3,4],[4,5]]

Output: 3

Explanation:

- Second 1: Nodes 1 and 5 disappear (each has 1 edge), edges [1,2] and [4,5] removed.
- Second 2: Nodes 2 and 4 now have 1 edge each, disappear, edges [2,3] and [3,4] removed.
- Second 3: Node 3 disappears.

Minimum time for full disappearance = 3

My thoughts: This is a perfect medium qns, can be solved using **topo-bfs** in graph.

A total of **89** students were shortlisted for the interview. The results were announced just one day before the interview giving me roughly 12 hours to cram a month's worth of prep.

Round 1 (Technical round 1):

The interview kicked off with the usual self-intro. As expected, he dove straight into my resume projects, skills, everything I had put down. The first part was basically a deep dive into my ML project:

- What does it do?
- How did I implement it?
- Which model did I use and why?
- Why did u choose this?
- And, of course, the results.

The interviewer didn't stop there; he went through all my other projects with the same level of curiosity. He also asked, how I implemented "*Interest based pair pooling*" which I have done in my project (like pairing some anonymous users to interact based on interest, similar to omegle but with filters). It was like a rapid-fire quiz: one question, one-line answer, bam, immediate follow-up. But somehow, I survived.

Then came a simple DSA question: **reverse the words in a sentence**. Easy right? I went from brute force to optimal solution, and he hit me with a follow-up: "*Now reverse the letters inside the words*". Literally twisting the whole string-reversal logic!! I answered that too.

Suddenly, the **Tech round felt more like a funny HR one**. He asked about my native place, MLA there, obviously we discussed about TVK and some politics. Later, He asked about **SAAS**, which I had on my resume. I ended up explaining **Techofes, Agni** in detail ... felt like a proud moment as an Organizer. We spent more time in this part.

Then context switching happened in real life HR qns again turned into Technical qns where it was left off, he threw in some **OS questions**: system calls, CPU scheduling, scheduling types and he asked me to explain **FCFS, Priority and Round Robin Scheduling** and other related concepts.. He also asked some cool things about **cpp**, asked me to explain about **pointers**, its advantages and its limitations. He also asked much more OS questions.

Finally, the round ended with a **puzzle**: I have three jugs => 5L, 8L, and 12L. How do I measure exactly 6L with minimum swap operations? At the end, the interviewer asked: "Do you have any questions for me?" I didn't ask him any. But U guys Don't skip it. Ask meaningful questions (about role, team culture, learning opportunities, etc.). It creates a positive impression.

After this round, I was shortlisted for the next stage. Out of all the participants, around 25 students were shortlisted for Technical Round 2.

Round-2:

This round had **two interviewers** — one virtual, one physical. The round started with me introduced myself. And then they asked

First questions:

- “Why CSE even though you were Bio-Math in 12th?”.
- “Do you like DBMS?” ...uh, really? How else do they expect me to answer? I said yes.

Then came **DBMS deep dive:**

- He asked, “*Have you backed up any DB data?*”, I clearly said no, still he questioned about **DBMS backup, then asked about sharding, horizontal vs vertical sharding, shard keys, how to shard and its usage** and lot questions on these two topics.
- Then asked normalization, anomalies, why we need it, all normal forms with examples. At 3rd normal form, he threw me a **mini-SQL task**: get student names studying CSE across student and department tables.
 - I did a **join query**; he tried to trick me: “*Why are you joining the tables? Now only you have split it! Still join requires? Then what is the need of that splitting*”
 - I calmly explained: **advantages and disadvantages of normalization** and also showed him a **subquery version** too.

Next, **OS:**

- He asked about **multithreading, parallelism, concurrency, multiprocessing** and plenty of questions as follow - ups.
- He asked, “**what would exactly happen at context switching**”, about **PCB** etc.,.
- He also asked, “*You have pc of single core, and u want to run a program which requires multiple core what your computer do?*”, I answered with **time-slicing, context switching, multiprocessing** also said “due to **absence of parallelism** the performance might be poor”.

Next, **DS:**

- He asked about **hashing**, Open addressing, closed addressing, chaining, linear & quadratic probing — basically my brain went on DSA overdrive.
- He asked, “*You have arr of size 10, but record of 10000, how do u hash and handle collision?*” I explained **chaining** approach for that.

- Then comes the classic: *"What's the last DS you learned?"* this is a tricky question, knowing he would questions in that, I still answered him as **Trie**, ended up explaining what is trie and what 's its real-world application and simulated a **search operation, auto completion** in trie.
- He also asked a **binary tree real-world example**.
- He asked me to Implement **Calculator using tree**. First, I have converted that **infix to postfix**, constructed the **expression tree**, where leaves are operands and the inner nodes are operators. And a simple modification of **postorder traversal** can do the magic for us! *"Avada Kedavra....."*.
- He also asked me about **heap** and told me to explain its underlying concepts and told to implement it.
- Later he gave me a **missing number problem** in range of 1 to n.

Next **OOPS** and **CPP** specific qns:

- He asked some questions about OOPS and he asked the classic **Diamond inheritance problem** and asked me *"How do I solve that?"* and he asked me to explain **virtual** keyword in detail. Also, he asked about **overriding** in both Java and CPP.
- Later they asked declaring array of size 5 and accessing arr[6] in both java and Cpp and asked, *"what will happen in both scenario?"*.

HR + casual chat:

- Later some HR questions ,here too... SAAS helped me to tackle some HR classical question like *"How do I resolve team conflict?"*.
- Found out the virtual guy was from **MIT**, He went on to share his college days experience and suddenly I realized this part of the interview was **longer than all the tough questions combined**.
- At the end, the interviewers asked: *"Do you have any questions?"*, I flipped the script and asked: *"Where do I see myself at SAP in 5 years?"*, they passionately answered (good guy, right?). I literally asked three to four questions as follow ups.
- And At last, I asked about **growth opportunities at SAP**, got some solid insights.

Fortunately, I was selected to the next round.

Round-3 HR Round:

Finally, the **most awaited HR round** happened. The guy was literally super chill. He started with the usual family background and some personal questions. Then came the classic: *"Why did you*

*switch from Bio-Math to CSE?" yep, **third time today**. I straight-up said, "Sir, this is the third time I'm answering this today!" He just smiled... and of course, he asked the same qns anyway.*

Then came the ultimate twister: He asked me, *"From all your life experience, give me an idea that will succeed **right now**, without any investment i.e no time, money, effort, or skills. That must work at that second at that instant."*, I asked what he meant as success in that qns .*He said, "Anything!"*. I was like *"Wait!... what kind of trap is this?"*, I tried some random answers, but he is not satisfied, At that point, my brain officially tried outsourced the question to ChatGPT, but unfortunately, no internet access in interviews. My brain threw a **404: Idea Not Found** error. no more ideas clicked, just little more frustration.... Anyway, that question basically **owned the entire HR round**. And at last, the round ended with me, completely drained and spitting out random ideas, just trying to escape from that trap.

Finally, I got selected into that company. **HURRAH!!!!**

Disclaimer:

All the above has been toned down to a professional version, removing the humorous elements of my interview. In reality, my interview was **much more humorous**. Honestly, it felt like all the rounds were HR-oriented, from the very first technical round to the last. I think, it is up to us to decide the flow and direction of the interview. I cracked many jokes in this interview and surprisingly that worked too.. Pls **Don't try this at home**, these kinds of stunts are depends entirely on interviewers and their personalities, Thank God I had some cool guys.

My Advice:

My advice to juniors is to approach interviews with a calm and confident mindset. Be chill, Don't panic, interviews aren't boss battles in **Soul Games**, they're just conversations (well, mostly). Avoid overthinking, and instead, focus on showcasing your passion and consistency, as both are key to long-term success . Be thorough with your resume and ensure you are well-prepared to discuss every detail you have included. Confidence, clarity, and composure will help you stand out. Finally, remember that not being selected is not the end there will always be more opportunities ahead. Choose your company wisely.

All the best Guys!!!

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